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# Operating Water Distribution Systems for Equitable Access to Clean Water †

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**Abstract:** Water distribution systems (WDSs) are designed to deliver potable water across urban areas. Unpredicted changes in water demands and hydraulics can increase the residence time in pipes, leading to the growth of microbes and decreased water quality at some locations in a network. During the COVID-19 pandemic, large-scale reductions in demands, especially in industrial and commercial areas as individuals worked from home, led to hot-spots of increased water age. In response to reduced water quality, consumers may avoid using tap water for end uses including drinking, cooking, and cleaning. The lack of access to clean water can create high costs for some households due to the cost of buying bottled water. Inequitable access to safe, affordable water is explored in this research in the context of the COVID-19 pandemic through a coupled framework. This research extends an existing agent-based modeling (ABM) framework that simulated COVID-19 transmission, social distancing decision-making, reductions in water demands, and flows in a water distribution system. The ABM is extended in this work to simulate households that perceive water quality problems with tap water and choose to buy bottled water for cooking, cleaning, and hygienic purposes. Agents choose tap water avoidance behaviors based on water age, a surrogate for water quality. Equity is evaluated using the cost of water, both tap and bottled, as a percentage of income. An optimization approach is coupled with the ABM framework and applied to design operational strategies that improve equitable access to safe affordable water. A graph theory approach identifies valves that should be opened and closed to improve water quality at nodes and maximize equity. The results demonstrate an increase in water age due to social distancing behaviors, and water of high age is observed to be disproportionately located near industrial areas. Adjusted income demonstrates inequities in access to safe and affordable water. Operational strategies are developed to improve equity for a community through valve operations that improve the equitable delivery of safe water. This research develops an approach to assess equity of the quality of delivered water and can be used to facilitate WDS management that provides equitable access to safe water.

**Keywords:** agent-based modeling; equity; clean water; COVID-19; bottled water; water age

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## 1. Introduction

Water distribution systems (WDSs) are designed to equitably deliver clean and safe water to all end users, yet many fall short due to changing conditions. Demands and flows in WDSs are dependent on decisions of end users to increase or decrease water use. Demand reductions cause increases in water age near the location of the demand change, leading to local water quality deterioration [1,2]. Previous work has shown that COVID-19 social distancing caused changes in demands that resulted in increased water age, primarily in residential areas close to industrial sites [3]. Low-income and minority households have

historically been located near industrial sites, raising equity concerns with respect to water quality. In response to reduced water quality, consumers may increase bottled water use, which leads to increased costs to access water for the end user. The complex interactions among water quality, bottled water buying decisions, and the hydraulic network supplying water requires a sociotechnical framework to capture the personal decision-making that is critical to the system.

The goal of this research is to develop a framework to assess the potentially inequitable water quality impacts of COVID-19 social distancing. Assessing equity is a primary goal of this research because past injustices such as the Flint water crisis have caused marginalized communities to lose trust in WDS operations and increase risk perceptions of the delivered water quality [4].

The framework developed in this research builds on previous work that simulated COVID-19 transmission, social distancing decision-making, and hydraulic system operation using an agent-based model (ABM) [3]. We build on this work by adding a tap water avoidance model using Bayesian belief network (BBN) models. Water age is quantitatively used to model agent perceptions of water quality, and risk perceptions are coupled with water quality to predict tap water avoidance. An equity metric is developed using the adjusted income of each agent, and pressure reducing valves are strategically placed to reduce inequities [5,6]. The new framework is applied for a case study to demonstrate the disparities in access to safe and affordable water and the effect of strategic infrastructure operations on improvements to equity.

## 2. Materials and Methods

This research develops an ABM framework to model inequities in delivered water quality. Operations are optimized to maximize equitable access to clean water. The ABM framework extends previous work that integrated a susceptible-exposed-infected-recovered (SEIR) COVID-19 transmission model, a BBN-driven social distancing model, and a hydraulic model [3]. This research includes a new cost of water framework to simulate tap water avoidance as a protective behavior when water quality degrades due to COVID-19 social distancing dynamics (Figure 1). Water age represents water quality based on stagnated water, and agents perceive the water quality water to make decisions to avoid tap water and buy bottled water. Households that buy bottled water use less tap water, further changing the hydraulics of the network including the water age. The cost of buying bottled water and tap water is modeled and compared to household income, using the cost of water as a percentage of income. New modules are described in the following sections, and the parameters that are included in the modules are listed in Table 1.

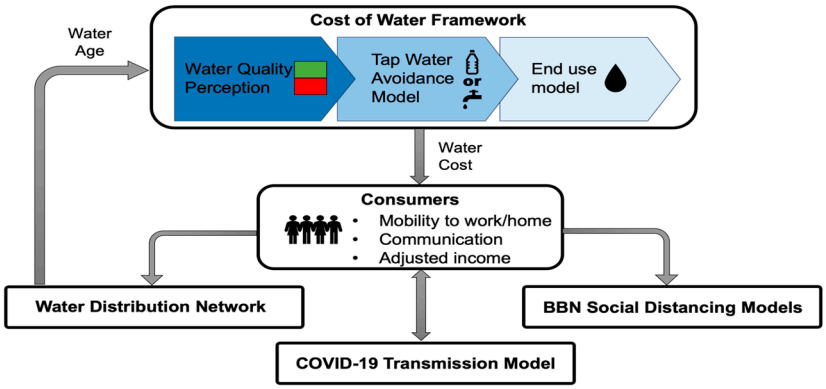


**Figure 1.** ABM framework integrates a new module to update the cost of water.

**Table 1.** ABM parameters are used to model tap water avoidance and adjusted income.

| Parameter | Symbol | Value |
|---|---|---|
| Bottled water buying decision | $D_i$ | [buy, don't buy] |
| Water cost | $C_W$ | Equation (2) |
| Cost of water as a percentage of income | $W_I$ | Equation (1) |

### 2.1. Cost of Water Framework

Household water cost is modeled using a framework that simulates agent perceptions of water quality, which informs agent decision-making on tap water avoidance behaviors and leads to changes in the cost of water. Households perceive water quality using the water age at their home node. Households use the water quality perceptions to decide to buy water for drinking, cooking, and hygiene use (e.g., face washing, teeth brushing). Buying bottled water reduces the demand on the hydraulic systems and increases household water expenditure.

### 2.2. Equity Metric: Cost of Water as a Percentage of Income

Equity is assessed using the cost of water as a percentage of income (Equation (1)).

$$W_I = \frac{C_W}{I_H}, \tag{1}$$

where $W_I$ is the cost of water as a percentage of income, $C_W$ is the cost of buying water, and $I_H$ is the household income. Each household is assigned an income value during model initialization. Households located closer to industrial sites have a higher chance of being assigned a lower income value to model historic siting inequities [7].

Households calculate the cost of water every thirty days (Equation (2)).

$$C_W = C_{TW} + C_{BW} \tag{2}$$

where $C_{TW}$ and $C_{BW}$ are the cost of purchasing tap water and bottled water, respectively. The cost used for bottled water is \$0.325/L, an average established by the International Bottled Water Association [8]. The cost of tap water is modeled using an increasing block structure.

At the end of the simulation, the cost of water as a percentage of income is calculated for each household (Equation (1)) and is used to analyze potential inequities caused by water age.

### 2.3. Valve Placement Optimization

A graph theory based algorithm is used to place pressure reducing valves (PRVs) throughout the system, using the cost of water as a percentage of income as an input [5]. PRVs are placed to reduce the pipe cost (see [5,6]), which includes the cost of water as a percentage of income. The disparities in water cost are reduced by strategically placed PRVs, which reduces the number of agents that adopt tap water avoidance behaviors.

## 3. Example Application

A virtual city, Micropolis, is used as the model community for this research and includes the hydraulic network. Micropolis was developed as a realistic water network used for WDS security simulations. There are 458 terminal nodes (434 residential, 15 industrial, and 9 commercial) and the total daily demand is 4.53 ML/day. Micropolis represents a small city with a population of 4606 agents modeled in the ABM.

## 4. Preliminary Results

Two scenarios were created to explore equity, a base case with no social distancing modeled and a prevention measure (PM) case with COVID-19 social distancing modeled. The PM case modeled working from home, dining out less, shopping for groceries less, and

wearing PPE using BBN models for agent decision-making. The cost of water, including both tap and bottled water, is reported for low- and median-income households for both the base case and PM case scenarios (Figure 2). In the base case, the difference between the low- and median-income households was small, suggesting that both groups face a similar cost of water burden. In the PM case, where social distancing is enabled, the difference is larger, suggesting that low-income households are bearing a larger cost of water burden. Results that further explore the impact of bottled water buying behaviors on water demands and equity will be reported in a conference presentation.

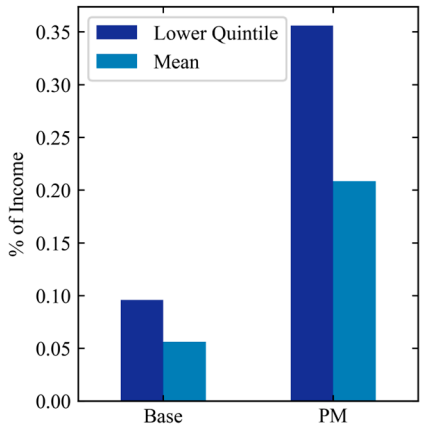


**Figure 2.** Cost of water as a percentage of household income for both base case and PM case.

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